Nandini Agrawal

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.19
Eminent Heights Public School	Class XII (MPBSE)	2024	83.40%
Trinity Convent School	Class X (CBSE)	2022	95.80%

PROJECTS

AttriSense -- Explainable Employee Attrition Risk Predictor | GitHub

Python | *Feature Engineering* | *EDA* | *XGBoost* | *SHAP* | *Streamlit*

Built an employee attrition prediction model using XGBoost on HR analytics data (~35 features), achieving **87% recall** on at-risk employees -- critical for minimizing missed flight-risk cases.

Improved model performance by 22% (PR-AUC) over a logistic regression baseline by **engineering domain-specific features** (tenure ratio, promotion gap, income-per-level) and addressing class imbalance via **SMOTE** and weighted loss functions.

Implemented SHAP-based explainability to surface individual- and department-level attrition drivers, translating model outputs into actionable retention insights for HR stakeholders (e.g., overtime, promotion stagnation).

SepsisSentinel -- Early-Warning Sepsis Prediction from ICU Vitals | GitHub

PyTorch | *GRU/LSTM* | *Pandas* | *Scikit-learn*

Built an end-to-end pipeline to preprocess hourly ICU vitals and labs (heart rate, respiration, temperature, blood pressure), handling missingness via **forward-fill** plus **missingness-indicator features** and constructing per-patient **sliding windows** for sequence modeling.

Trained a **bidirectional GRU** over multivariate ICU time-series data from **8,000** patients, reframing sepsis detection as a **6-hour-ahead** early-warning problem instead of point-in-time classification.

DeadlockGuard -- Deadlock Avoidance & Detection Engine | GitHub

C++ | *STL* | *Banker's Algorithm* | *Graph Algorithms*

Engineered a C++ command-line tool demonstrating classical OS deadlock handling, implementing the Banker's Algorithm for deadlock avoidance and DFS-based wait-for graph cycle detection for deadlock detection, with a menu-driven CLI supporting file-based and manual input.

Built a safe-state analysis engine that computes valid safe sequences and simulates resource requests, applying tentative allocation with automatic rollback on unsafe states

SKILLS

Languages: C, C++, HTML, CSS, Python, SQL

ML Frameworks & Tools: NumPy, Pandas, Matplotlib, PyTorch, Git, GitHub

CS Fundamentals: Object Oriented Programming (OOP), Data Structures, Algorithms, Operating Systems

Soft Skills: Leadership, Teamwork, Communication, Problem Solving

RELEVANT COURSES & CERTIFICATIONS

CS/ AI: Intro. to Programming, Intro. to AI, Probability, Elementary Linear Algebra, Calculus

Machine Learning Specialization by Andrew Ng (Coursera)

Supervised Machine Learning, Advanced Learning Algorithms, Unsupervised Learning, Recommenders

Deep Learning Specialization by Andrew Ng (Coursera)

Neural Networks, Hyperparameter Tuning, Regularization, Optimization, CNNs, Sequence Models

POSITIONS OF RESPONSIBILITY

Internship Coordinator - *Office of Career Services*

Coordinated **end-to-end internship recruitment drives**, managing recruiter communication and student applications.

Streamlined **candidature verification** and tracking processes, **ensuring smooth execution** during recruitment drives.

Content and Speaker Curator - *Extramural Lectures*

Curated and executed **high-profile guest lectures** and panel discussions by coordinating directly with industry leaders, academicians, and distinguished speakers, managing all end-to-end event logistics, scheduling, and speaker hospitality.

Led publicity and outreach initiatives that increased **student participation** and engagement across lecture sessions.

Core Member - *Rang de Manch (Dramatics Club)*

Orchestrated the production, scriptwriting, and performance execution of stage and street play, representing IIT Hyderabad at **Inter-IIT Cultural Meets 7.0 & 8.0**.

Mentored and trained **14+** freshers in acting, public speaking, and stage performance, successfully preparing them for institute-wide theatrical showcases.

EXTRACURRICULARS

Played a pivotal role in the **Street Play, Stage Play, Mime** at the **Inter-IIT Cultural Meets 7.0 & 8.0.**

Led the **end-to-end preparation** of the Freshers' Stage Play by driving rehearsals and mentoring performers.